Predictive Modeling for Car Price Estimation:

Abstract:

The automotive industry is constantly evolving, and the ability to accurately predict the price of a car plays a crucial role in various domains such as sales, insurance, and finance. This research focuses on developing a robust predictive model for car price estimation. The study leverages machine learning techniques, specifically linear regression, to analyze and model the relationships between key features and the market value of automobiles.

The dataset used in this research encompasses a diverse range of car attributes, including make, model, year, mileage, fuel type, and other relevant factors. Data preprocessing techniques, such as handling missing values, encoding categorical variables, and scaling numerical features, are applied to ensure the quality and consistency of the dataset.

A predictive model is constructed using scikit-learn, a popular machine learning library in Python. The linear regression algorithm is employed to capture the underlying patterns within the data and establish a correlation between the input features and the target variable, i.e., the car price. The model is trained on a subset of the dataset and validated using a separate test set to assess its generalization capabilities.

The evaluation metrics, including the R-squared score, are employed to quantify the model's accuracy and effectiveness in predicting car prices. The study also explores the impact of various features on the predictive performance and identifies the most influential factors in determining car prices.

The developed predictive model holds the potential to provide valuable insights for pricing strategies, enabling businesses and consumers to make informed decisions. Additionally, the research contributes to the broader field of data-driven decision-making within the automotive industry. The findings and methodologies outlined in this study serve as a foundation for future enhancements and applications in related domains.

Here's how the model is made -

```
from google.colab import files
data_to_load = files.upload()
```

Choose Files quikr_car.csv

- **quikr_car.csv**(text/csv) - 60094 bytes, last modified: 2/24/2024 - 100% done

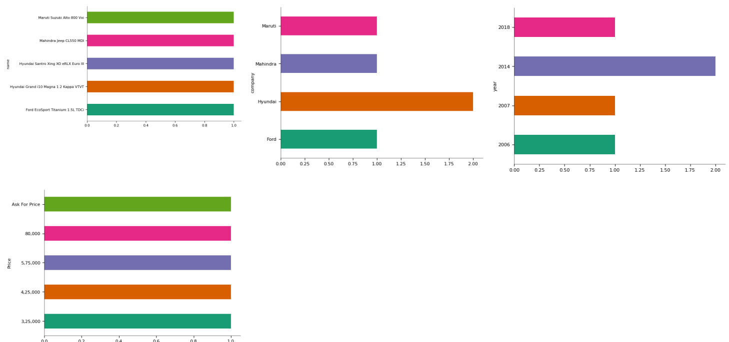
```
Saving quikr_car.csv to quikr_car (1).csv
```

```
#importing necessary libraries that are going to be use
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import make_column_transformer
from sklearn.pipeline import make_pipeline
```

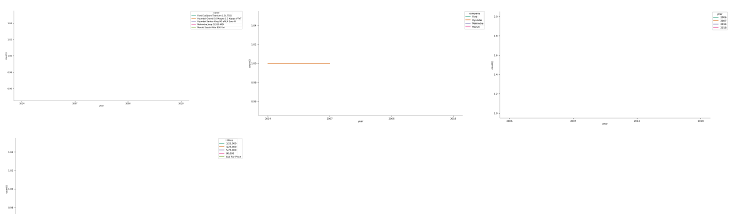
```
car=pd.read_csv('quikr_car.csv')
car.head()
```

	name	company	year	Price	kms_driven	fuel_type
0	Hyundai Santro Xing XO eRLX Euro III	Hyundai	2007	80,000	45,000 kms	Petrol
1	Mahindra Jeep CL550 MDI	Mahindra	2006	4,25,000	40 kms	Diesel
2	Maruti Suzuki Alto 800 Vxi	Maruti	2018	Ask For Price	22,000 kms	Petrol
3	Hyundai Grand i10 Magna 1.2 Kappa VTVT	Hyundai	2014	3,25,000	28,000 kms	Petrol
4	Ford EcoSport Titanium 1.5L TDCi	Ford	2014	5,75,000	36,000 kms	Diesel

Categorical distributions



Time series



Next steps: Generate code with car View recommended plots

```
car.shape
```

```
(892, 6)
```

```
car.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 892 entries, 0 to 891
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   name        892 non-null    object
 1   company     892 non-null    object
 2   year        892 non-null    object
 3   Price       892 non-null    object
 4   kms_driven  840 non-null    object
 5   fuel_type   837 non-null    object
dtypes: object(6)
memory usage: 41.9+ KB
```

Checking the uncleaned data

```
car['year'].unique()
```

```
array(['2007', '2006', '2018', '2014', '2015', '2012', '2013', '2016',
       '2010', '2017', '2008', '2011', '2019', '2009', '2005', '2000',
       '...', '150k', 'TOUR', '2003', 'r 15', '2004', 'Zest', '/-Rs',
       'sale', '1995', 'ara)', '2002', 'SELL', '2001', 'tion', 'odel',
```

```
       '2 bs', 'arry', 'Eon', 'o...', 'ture', 'emi', 'car', 'able', 'no.',
       'd...', 'SALE', 'digo', 'sell', 'd Ex', 'n...', 'e...', 'D...',
       ', Ac', 'go .', 'k...', 'o c4', 'zire', 'cent', 'Sumo', 'cab',
       't xe', 'EV2', 'r...', 'zest'], dtype=object)
```

```
car['Price'].unique()
```

```
array(['80,000', '4,25,000', 'Ask For Price', '3,25,000', '5,75,000',
       '1,75,000', '1,90,000', '8,30,000', '2,50,000', '1,82,000',
       '3,15,000', '4,15,000', '3,20,000', '10,00,000', '5,00,000',
       '3,50,000', '1,60,000', '3,10,000', '75,000', '1,00,000',
       '2,90,000', '95,000', '1,80,000', '3,85,000', '1,05,000',
       '6,50,000', '6,89,999', '4,48,000', '5,49,000', '5,01,000',
       '4,89,999', '2,80,000', '3,49,999', '2,84,999', '3,45,000',
       '4,99,999', '2,35,000', '2,49,999', '14,75,000', '3,95,000',
       '2,20,000', '1,70,000', '85,000', '2,00,000', '5,70,000',
       '1,10,000', '4,48,999', '18,91,111', '1,59,500', '3,44,999',
       '4,49,999', '8,65,000', '6,99,000', '3,75,000', '2,24,999',
       '12,00,000', '1,95,000', '3,51,000', '2,40,000', '90,000',
       '1,55,000', '6,00,000', '1,89,500', '2,10,000', '3,90,000',
       '1,35,000', '16,00,000', '7,01,000', '2,65,000', '5,25,000',
       '3,72,000', '6,35,000', '5,50,000', '4,85,000', '3,29,500',
       '2,51,111', '5,69,999', '69,999', '2,99,999', '3,99,999',
       '4,50,000', '2,70,000', '1,58,400', '1,79,000', '1,25,000',
       '2,99,000', '1,50,000', '2,75,000', '2,85,000', '3,40,000',
       '70,000', '2,89,999', '8,49,999', '7,49,999', '2,74,999',
       '9,84,999', '5,99,999', '2,44,999', '4,74,999', '2,45,000',
       '1,69,500', '3,70,000', '1,68,000', '1,45,000', '98,500',
       '2,09,000', '1,85,000', '9,00,000', '6,99,999', '1,99,999',
       '5,44,999', '1,99,000', '5,40,000', '49,000', '7,00,000', '55,000',
       '8,95,000', '3,55,000', '5,65,000', '3,65,000', '40,000',
       '4,00,000', '3,30,000', '5,80,000', '3,79,000', '2,19,000',
       '5,19,000', '7,30,000', '20,00,000', '21,00,000', '14,00,000',
       '3,11,000', '8,55,000', '5,35,000', '1,78,000', '3,00,000',
       '2,55,000', '5,49,999', '3,80,000', '57,000', '4,10,000',
       '2,25,000', '1,20,000', '59,000', '5,99,000', '6,75,000', '72,500',
       '6,10,000', '2,30,000', '5,20,000', '5,24,999', '4,24,999',
       '6,44,999', '5,84,999', '7,99,999', '4,44,999', '6,49,999',
       '9,44,999', '5,74,999', '3,74,999', '1,30,000', '4,01,000',
       '13,50,000', '1,74,999', '2,39,999', '99,999', '3,24,999',
       '10,74,999', '11,30,000', '1,49,000', '7,70,000', '30,000',
       '3,35,000', '3,99,000', '65,000', '1,69,999', '1,65,000',
       '5,60,000', '9,50,000', '7,15,000', '45,000', '9,40,000',
       '1,55,555', '15,00,000', '4,95,000', '8,00,000', '12,99,000',
       '5,30,000', '14,99,000', '32,000', '4,05,000', '7,60,000',
       '7,50,000', '4,19,000', '1,40,000', '15,40,000', '1,23,000',
       '4,98,000', '4,80,000', '4,88,000', '15,25,000', '5,48,900',
       '7,25,000', '99,000', '52,000', '28,00,000', '4,99,000',
       '3,81,000', '2,78,000', '6,90,000', '2,60,000', '90,001',
       '1,15,000', '15,99,000', '1,59,000', '51,999', '2,15,000',
       '35,000', '11,50,000', '2,69,000', '60,000', '4,30,000',
       '85,00,003', '4,01,919', '4,90,000', '4,24,000', '2,05,000',
       '5,49,900', '3,71,500', '4,35,000', '1,89,700', '3,89,700',
       '3,60,000', '2,95,000', '1,14,990', '10,65,000', '4,70,000',
       '48,000', '1,88,000', '4,65,000', '1,79,999', '21,90,000',
       '23,90,000', '10,75,000', '4,75,000', '10,25,000', '6,15,000',
       '19,00,000', '14,90,000', '15,10,000', '18,50,000', '7,90,000',
       '17,25,000', '12,25,000', '68,000', '9,70,000', '31,00,000',
       '8,99,000', '88,000', '53,000', '5,68,500', '71,000', '5,90,000',
       '7,95,000', '42,000', '1,89,000', '1,62,000', '35,999',
       '29,00,000', '39,999', '50,500', '5,10,000', '8,60,000',
       '5,00,001'], dtype=object)
```

```
car['kms_driven'].unique()
```

```
       '4,000 kms', '16,934 kms', '43,000 kms', '35,550 kms',
       '39,522 kms', '39,000 kms', '55,000 kms', '72,000 kms',
       '15,975 kms', '70,000 kms', '23,452 kms', '35,522 kms',
       '48,508 kms', '15,487 kms', '82,000 kms', '20,000 kms',
       '68,000 kms', '38,000 kms', '27,000 kms', '33,000 kms',
       '46,000 kms', '16,000 kms', '47,000 kms', '35,000 kms',
       '30,874 kms', '15,000 kms', '29,685 kms', '1,30,000 kms',
       '19,000 kms', nan, '54,000 kms', '13,000 kms', '38,200 kms',
       '50,000 kms', '13,500 kms', '3,600 kms', '45,863 kms',
       '60,500 kms', '12,500 kms', '18,000 kms', '13,349 kms',
```

```
       '24,330 kms', '65,480 kms', '28,028 kms', '2,00,000 kms',
       '99,000 kms', '2,800 kms', '21,000 kms', '11,000 kms',
       '66,000 kms', '3,000 kms', '7,000 kms', '38,500 kms', '37,200 kms',
       '43,200 kms', '24,800 kms', '45,872 kms', '40,000 kms',
       '11,400 kms', '97,200 kms', '52,000 kms', '31,000 kms',
       '1,75,430 kms', '37,000 kms', '65,000 kms', '3,350 kms',
       '75,000 kms', '62,000 kms', '73,000 kms', '2,200 kms',
       '54,870 kms', '34,580 kms', '97,000 kms', '60 kms', '80,200 kms',
       '3,200 kms', '0,000 kms', '5,000 kms', '588 kms', '71,200 kms',
       '1,75,400 kms', '9,300 kms', '56,758 kms', '10,000 kms',
       '56,450 kms', '56,000 kms', '32,700 kms', '9,000 kms', '73 kms',
       '1,60,000 kms', '84,000 kms', '58,559 kms', '57,000 kms',
       '1,70,000 kms', '80,000 kms', '6,821 kms', '23,000 kms',
       '34,000 kms', '1,800 kms', '4,00,000 kms', '48,000 kms',
       '90,000 kms', '12,000 kms', '69,900 kms', '1,66,000 kms',
       '122 kms', '0 kms', '24,000 kms', '36,469 kms', '7,800 kms',
       '24,695 kms', '15,141 kms', '59,910 kms', '1,00,000 kms',
       '4,500 kms', '1,29,000 kms', '300 kms', '1,31,000 kms',
       '1,11,111 kms', '59,466 kms', '25,500 kms', '44,005 kms',
       '2,110 kms', '43,222 kms', '1,00,200 kms', '65 kms',
       '1,40,000 kms', '1,03,553 kms', '58,000 kms', '1,20,000 kms',
       '49,800 kms', '100 kms', '81,876 kms', '6,020 kms', '55,700 kms',
       '18,500 kms', '1,80,000 kms', '53,000 kms', '35,500 kms',
       '22,134 kms', '1,000 kms', '8,500 kms', '87,000 kms', '6,000 kms',
       '15,574 kms', '8,000 kms', '55,800 kms', '56,400 kms',
       '72,160 kms', '11,500 kms', '1,33,000 kms', '2,000 kms',
       '88,000 kms', '65,422 kms', '1,17,000 kms', '1,50,000 kms',
       '10,750 kms', '6,800 kms', '5 kms', '9,800 kms', '57,923 kms',
       '30,201 kms', '6,200 kms', '37,518 kms', '24,652 kms', '383 kms',
       '95,000 kms', '3,528 kms', '52,500 kms', '47,900 kms',
       '52,800 kms', '1,95,000 kms', '48,008 kms', '48,247 kms',
       '9,400 kms', '64,000 kms', '2,137 kms', '10,544 kms', '49,500 kms',
       '1,47,000 kms', '90,001 kms', '48,006 kms', '74,000 kms',
       '85,000 kms', '29,500 kms', '39,700 kms', '67,000 kms',
       '19,336 kms', '60,105 kms', '45,933 kms', '1,02,563 kms',
       '28,600 kms', '41,800 kms', '1,16,000 kms', '42,590 kms',
       '7,400 kms', '54,500 kms', '76,000 kms', '00 kms', '11,523 kms',
       '38,600 kms', '95,500 kms', '37,458 kms', '85,960 kms',
       '12,516 kms', '30,600 kms', '2,550 kms', '62,500 kms',
       '69,000 kms', '28,400 kms', '68,485 kms', '3,500 kms',
       '85,455 kms', '63,000 kms', '1,600 kms', '77,000 kms',
       '26,500 kms', '2,875 kms', '13,900 kms', '1,500 kms', '2,450 kms',
       '1,625 kms', '33,400 kms', '60,123 kms', '38,900 kms',
       '1,37,495 kms', '91,200 kms', '1,46,000 kms', '1,00,800 kms',
       '2,100 kms', '2,500 kms', '1,32,000 kms', 'Petrol'], dtype=object)
```

```
car['fuel_type'].unique()
```

```
array(['Petrol', 'Diesel', nan, 'LPG'], dtype=object)
```

Cleaning

```
backup=car.copy()
```

```
# Ensure 'year' column is treated as string before checking for numeric values
car = car[car['year'].str.isnumeric()]

# Convert 'kms_driven' to integer type
car['year'] = car['year'].astype(int)

# Display the 'year' column for the filtered DataFrame
print(car['year'])
```

```
0      2007
1      2006
2      2018
3      2014
4      2014
       ...
886    2009
888    2018
889    2013
890    2014
```

```
891    2014
Name: year, Length: 842, dtype: int64
```

```
# Convert 'Price' column to string and remove commas
car['Price'] = car['Price'].astype(str).str.replace(',', '')

# Filter out rows where 'Price' is "Ask For Price"
car = car[car['Price'] != "Ask For Price"]

car['Price'] = car['Price'].astype(int)

# Display the 'Price' column for the filtered DataFrame
print(car['Price'])
```

```
0       80000
1      425000
3      325000
4      575000
6      175000
        ...
886    300000
888    260000
889    390000
890    180000
891    160000
Name: Price, Length: 819, dtype: int64
<ipython-input-63-12fa726ba3ef>:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-vie
  car['Price'] = car['Price'].astype(int)
```

```
# Convert 'kms_driven' column to string type
car['kms_driven'] = car['kms_driven'].astype(str)

# Split, extract, and remove commas from 'kms_driven' column
car['kms_driven'] = car['kms_driven'].str.split(' ').str.get(0).str.replace(',', '')

# Filter out rows where 'kms_driven' is not numeric
car = car[car['kms_driven'].str.isnumeric()]

# Convert 'kms_driven' to integer type
car['kms_driven'] = car['kms_driven'].astype(int)

# Display the resulting DataFrame
print(car['kms_driven'])
```

```
0       45000
1          40
3       28000
4       36000
6       41000
        ...
883     50000
885     30000
886    132000
888     27000
889     40000
Name: kms_driven, Length: 817, dtype: int64
<ipython-input-64-8b58d278bced>:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-vie
  car['kms_driven'] = car['kms_driven'].astype(str)
<ipython-input-64-8b58d278bced>:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-vie
  car['kms_driven'] = car['kms_driven'].str.split(' ').str.get(0).str.replace(',', '')
```

```
# filters out rows where the 'fuel_type' column is NaN
car=car[~car['fuel_type'].isna()]

# Display the resulting DataFrame
print(car['fuel_type'])
```

```
0      Petrol
1      Diesel
3      Petrol
4      Diesel
6      Diesel
        ...
883    Petrol
885    Diesel
886    Petrol
888    Diesel
889    Diesel
Name: fuel_type, Length: 816, dtype: object
```

```
car['name']=car['name'].str.split(' ').str.slice(0,3).str.join(' ')
print(car['name'])
```

```
0         Hyundai Santro Xing
1         Mahindra Jeep CL550
3           Hyundai Grand i10
4      Ford EcoSport Titanium
6                   Ford Figo
                ...
883        Maruti Suzuki Ritz
885            Tata Indica V2
886      Toyota Corolla Altis
888              Tata Zest XM
889        Mahindra Quanto C8
Name: name, Length: 816, dtype: object
```

cleaned data

```
car=car.reset_index(drop=True)
print(car)
```

```
                       name   company  year   Price  kms_driven fuel_type
0       Hyundai Santro Xing   Hyundai  2007   80000       45000    Petrol
1       Mahindra Jeep CL550  Mahindra  2006  425000          40    Diesel
2         Hyundai Grand i10   Hyundai  2014  325000       28000    Petrol
3    Ford EcoSport Titanium      Ford  2014  575000       36000    Diesel
4                 Ford Figo      Ford  2012  175000       41000    Diesel
..                      ...       ...   ...     ...         ...       ...
811      Maruti Suzuki Ritz    Maruti  2011  270000       50000    Petrol
812          Tata Indica V2      Tata  2009  110000       30000    Diesel
813    Toyota Corolla Altis    Toyota  2009  300000      132000    Petrol
814            Tata Zest XM      Tata  2018  260000       27000    Diesel
815      Mahindra Quanto C8  Mahindra  2013  390000       40000    Diesel

[816 rows x 6 columns]
```

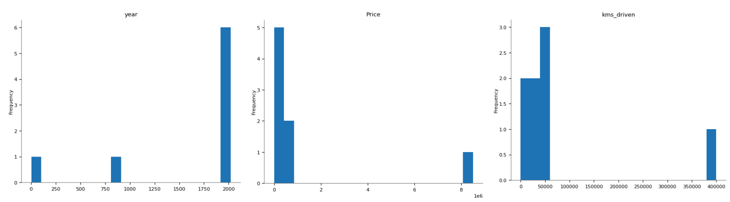
```
car.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 816 entries, 0 to 815
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   name        816 non-null    object
 1   company     816 non-null    object
 2   year        816 non-null    int64 
 3   Price       816 non-null    int64 
 4   kms_driven  816 non-null    int64 
 5   fuel_type   816 non-null    object
dtypes: int64(3), object(3)
memory usage: 38.4+ KB
```

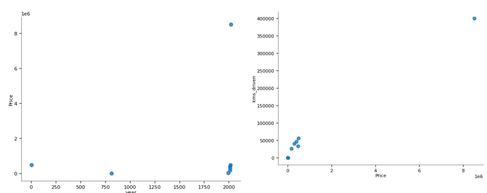
```
car.describe()
```

	year	Price	kms_driven
count	816.000000	8.160000e+02	816.000000
mean	2012.444853	4.117176e+05	46275.531863
std	4.002992	4.751844e+05	34297.428044
min	1995.000000	3.000000e+04	0.000000
25%	2010.000000	1.750000e+05	27000.000000
50%	2013.000000	2.999990e+05	41000.000000
75%	2015.000000	4.912500e+05	56818.500000
max	2019.000000	8.500003e+06	400000.000000

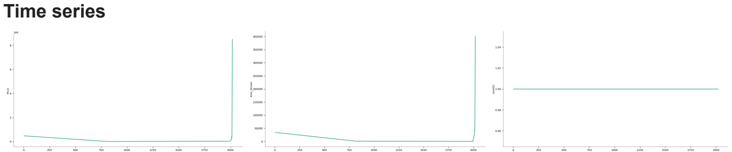
Distributions



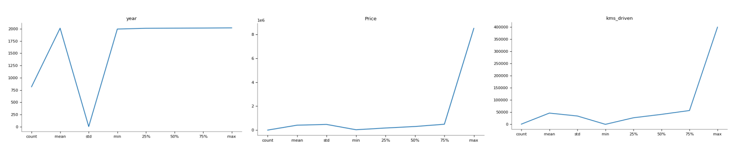
2-d distributions



Time series



Values



```
car = car[car['Price']<6e6].reset_index(drop=True)
print(car)
```

```
                        name   company  year   Price  kms_driven fuel_type
0        Hyundai Santro Xing   Hyundai  2007   80000       45000    Petrol
1        Mahindra Jeep CL550  Mahindra  2006  425000          40    Diesel
2          Hyundai Grand i10   Hyundai  2014  325000       28000    Petrol
3     Ford EcoSport Titanium      Ford  2014  575000       36000    Diesel
4                  Ford Figo      Ford  2012  175000       41000    Diesel
..                       ...       ...   ...     ...         ...       ...
810       Maruti Suzuki Ritz    Maruti  2011  270000       50000    Petrol
811           Tata Indica V2      Tata  2009  110000       30000    Diesel
812     Toyota Corolla Altis    Toyota  2009  300000      132000    Petrol
813             Tata Zest XM      Tata  2018  260000       27000    Diesel
814       Mahindra Quanto C8  Mahindra  2013  390000       40000    Diesel

[815 rows x 6 columns]
```

New cleaned dataset

```
car.to_csv('Cleaned Car.csv')
```

Model

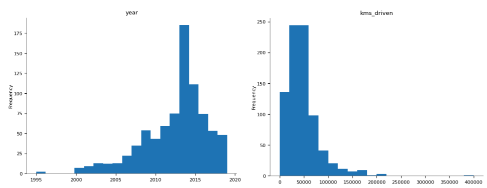
```
X = car.drop(columns='Price')
y = car['Price']
```

```
#print X
X
```

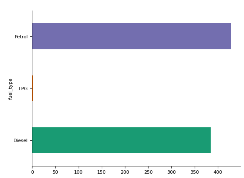
	name	company	year	kms_driven	fuel_type
0	Hyundai Santro Xing	Hyundai	2007	45000	Petrol
1	Mahindra Jeep CL550	Mahindra	2006	40	Diesel
2	Hyundai Grand i10	Hyundai	2014	28000	Petrol
3	Ford EcoSport Titanium	Ford	2014	36000	Diesel
4	Ford Figo	Ford	2012	41000	Diesel
...	...	...	...	...	...
810	Maruti Suzuki Ritz	Maruti	2011	50000	Petrol
811	Tata Indica V2	Tata	2009	30000	Diesel
812	Toyota Corolla Altis	Toyota	2009	132000	Petrol
813	Tata Zest XM	Tata	2018	27000	Diesel
814	Mahindra Quanto C8	Mahindra	2013	40000	Diesel

815 rows × 5 columns

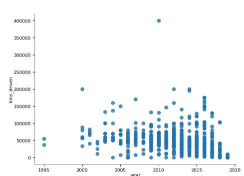
Distributions



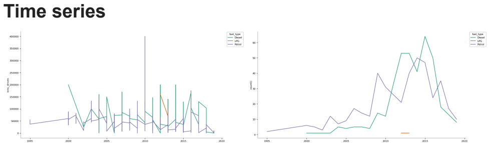
Categorical distributions



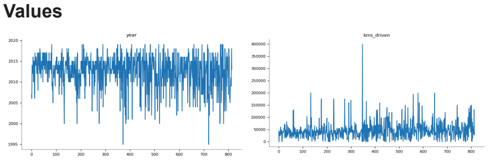
2-d distributions



Time series



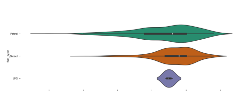
Values



Faceted distributions

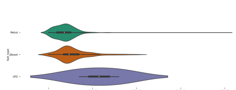
```
<string>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and
```



```
<string>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and
```



```
# print y
y
```

```
0       80000
1      425000
2      325000
3      575000
4      175000
        ...
810    270000
811    110000
812    300000
813    260000
814    390000
Name: Price, Length: 815, dtype: int64
```

```
# from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size = 0.2)
```

```
print(X.shape, X_train.shape, X_test.shape)
```

```
(815, 5) (652, 5) (163, 5)
```

```
# from sklearn.preprocessing import OneHotEncoder
ohe = OneHotEncoder()
ohe.fit(X[['name', 'company', 'fuel_type']])
```

```
▾ OneHotEncoder
OneHotEncoder()
```

```
ohe.categories_
```

```
             'Land Rover Freelander', 'Mahindra Bolero DI',
             'Mahindra Bolero Power', 'Mahindra Bolero SLE',
             'Mahindra Jeep CL550', 'Mahindra Jeep MM', 'Mahindra KUV100',
             'Mahindra KUV100 K8', 'Mahindra Logan', 'Mahindra Logan Diesel',
             'Mahindra Quanto C4', 'Mahindra Quanto C8', 'Mahindra Scorpio',
```

```
           'Tata Zest Quadrajet', 'Tata Zest XE', 'Tata Zest XM',
           'Toyota Corolla', 'Toyota Corolla Altis', 'Toyota Corolla H2',
           'Toyota Etios', 'Toyota Etios G', 'Toyota Etios GD',
           'Toyota Etios Liva', 'Toyota Fortuner', 'Toyota Fortuner 3.0',
           'Toyota Innova 2.0', 'Toyota Innova 2.5', 'Toyota Qualis',
           'Volkswagen Jetta Comfortline', 'Volkswagen Jetta Highline',
           'Volkswagen Passat Diesel', 'Volkswagen Polo',
           'Volkswagen Polo Comfortline', 'Volkswagen Polo Highline',
           'Volkswagen Polo Highline1.2L', 'Volkswagen Polo Trendline',
           'Volkswagen Vento Comfortline', 'Volkswagen Vento Highline',
           'Volkswagen Vento Konekt', 'Volvo S80 Summum'], dtype=object),
    array(['Audi', 'BMW', 'Chevrolet', 'Datsun', 'Fiat', 'Force', 'Ford',
           'Hindustan', 'Honda', 'Hyundai', 'Jaguar', 'Jeep', 'Land',
           'Mahindra', 'Maruti', 'Mercedes', 'Mini', 'Mitsubishi', 'Nissan',
           'Renault', 'Skoda', 'Tata', 'Toyota', 'Volkswagen', 'Volvo'],
          dtype=object),
    array(['Diesel', 'LPG', 'Petrol'], dtype=object)]
```

```
# from sklearn.compose import make_column_transformer
column_trans = make_column_transformer((OneHotEncoder(categories=ohe.categories_), ['name', 'company', 'fuel_type']), remainder
```

```
# from sklearn.linear_model import LinearRegression
lr = LinearRegression()
```

```
# from sklearn.pipeline import make_pipeline
pipe = make_pipeline(column_trans, lr)
```

```
pipe.fit(X_train, y_train)
```

```
Pipeline
columntransformer: ColumnTransformer
onehotencoder    remainder
OneHotEncoder    passthrough
                 passthrough
LinearRegression
LinearRegression()
```

```
y_pred = pipe.predict(X_test)
y_pred
```

```
array([ 297457.40176061,  570215.91927182,  541841.6531882 ,
        516851.10788282,  362646.7365185 ,  209846.58662486,
        575761.49544433,  220414.06789605,  856320.87652411,
        472448.77618982,  -12355.30690742,  311566.80537783,
        522630.01359042,   34342.47274411,  192967.08504654,
        388259.34757585,   77470.44331238,  286015.66559072,
        129225.37339483,  458618.27822303,   15405.04379655,
       1130209.31377405,  490337.85781549,  -25112.30719891,
        199560.16834357,  413879.12468081,   80194.98796228,
        240681.10094978,  414295.21667203,  231053.23509706,
        341061.22878225,  397310.24705131,  835620.1208956 ,
        230102.62846632,  -28248.13768172,  116232.51954482,
        670058.18308564,  204394.78189785,  635380.60805061,
        467862.68796055,  153310.37076075,  218528.24282032,
        189532.14986221,  336610.19877282,  136347.90078178,
        467862.68796055,  632689.87858711,  174706.57172226,
        312238.24403944,  319889.07086171,  211246.03108994,
        291629.45084523,  363998.30750785,  541478.38211747,
        151780.30459882,  195095.65263877,  297557.66489102,
        223100.50398418,  250527.27006502,  103275.25163731,
        657248.15405568,  819356.84740672,  297106.48080415,
        323979.06254418,  295073.25785483,  456676.87571767,
       1704423.48112448,  449660.50452796,  203696.49878135,
        507432.7147925 ,  360179.78169296,  576151.63025658,
        583311.85851573,  -97648.5710584 ,  269311.80636044,
        356437.90912838,  128700.54179306,  681144.94729403,
        305936.91684128,  622343.64245877,  464927.0617239 ,
        492457.88873711, 1006428.05480178, 1080712.59615217,
        307904.7012046 ,  232914.17463063,  657248.15405568,
        438400.8890816 ,  541379.94570476,  298760.822456  ,
```

```
       285414.08680823,   83111.08775194,   70226.48961563,
       433439.11239561,  192604.66510487,  283201.74421154,
       448196.95737093,  246182.14831734,  991533.04751989,
       146773.35929281,  481251.56307936,  300964.26416778,
       239396.93310621,  496999.45091886,  565244.3054463 ,
         5829.52072459, -181573.49989252,  301743.64232226,
       123741.27466035,  236596.44200997,  510538.7095294 ,
        47957.41751671,  124541.82348866,  566820.52118097,
       281194.85861631,  369038.22625112,  507544.3639602 ,
       398362.44391759,  354345.57694643,  143976.48229045,
       335682.2351009 ,  457050.09390853,  477439.69642968,
        78776.21677808,    7945.7850734 ,  617220.61136998,
       152538.09144858,  278888.12007349,  535830.32902809,
       647971.19585255,  388897.03893081,  525183.71413822,
       465680.87481899,  316195.68390947,  364650.01785555,
       395385.0403415 ,  153210.10763033,  377489.02336669,
       461812.59260324,  971331.60381383,  310195.71373458,
      1111775.45378231,  329405.23523857,  269400.97403252,
       209508.19666979,  458361.19970378,  253425.08813201,
       514373.78547166,  157209.11035337,  437012.47624551,
       299863.71689057,  309514.13687976,  252390.91128749,
       432916.56850741,  538507.78077658,  240336.90184377,
        47121.37657417,  140331.8507992 ,  394289.15857965,
       835493.88961441,  291629.45084523,  155887.76183027,
       430449.85085632])
```

```
from sklearn.metrics import r2_score
```

```
r2_score(y_test, y_pred)
```

```
0.8456515104452564
```

```
scores = []
for i in range(1000):
  X_train, X_test, y_train, y_test = train_test_split(X,y, test_size = 0.2, random_state=i)
  lr = LinearRegression()
  pipe = make_pipeline(column_trans, lr)
  pipe.fit(X_train, y_train)
  y_pred = pipe.predict(X_test)
  scores.append(r2_score(y_test, y_pred))
```

```
np.argmax(scores)
```

```
433
```

```
# extracting the maximum score from the array
scores[np.argmax(scores)]
```

```
0.8456515104452564
```

```
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size = 0.2, random_state=np.argmax(scores))
lr = LinearRegression()
pipe = make_pipeline(column_trans, lr)
pipe.fit(X_train, y_train)
y_pred = pipe.predict(X_test)
r2_score(y_test, y_pred)
```

```
0.8456515104452564
```

```
import pickle
```

```
pickle.dump(pipe, open('LinearRegressionModel.pkl', 'wb'))
```

Model is ready to predict

```
pipe.predict(pd.DataFrame([['Maruti Suzuki Swift', 'Maruti', 2019, 100, 'Petrol']], columns=['name', 'company', 'year', 'kms_dr:
```

```
array([459113.49353657])
```